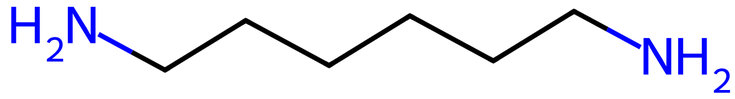
A shared fold can hide different substrate chemistry

ENZYME	NATIVE NITROGEN DONOR
GS	NH_3
GmaS	
PuuA	

TARGET AMINE

Hexamethylenediamine



A diamine-compatible lead

Search decision: investigate the PuuA-related lead

Native N-donor examples: IUBMB EC 6.3.1.2 / 6.3.4.12; Kurihara et al., 2008. Target-substrate activity remains unestablished.

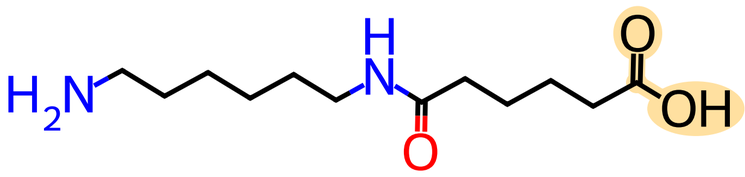
DesD: the same substrates give different products

DONOR

ACCEPTOR

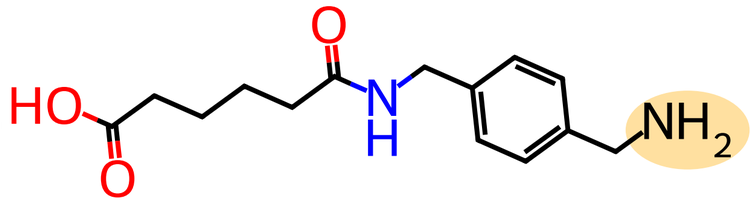
PRODUCT

MA



+

XA

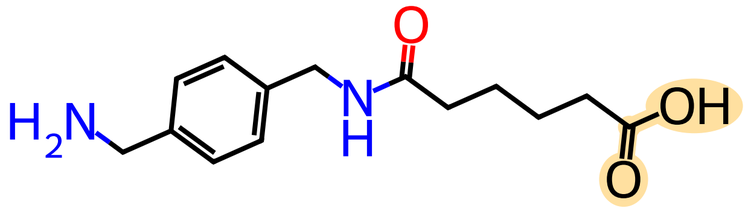


→

M A X A

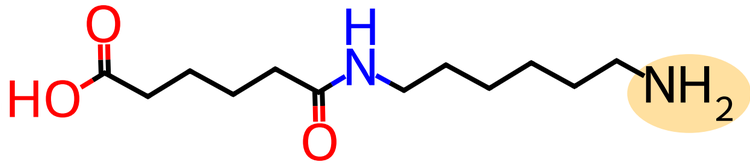
Higher yield

XA



+

MA



→

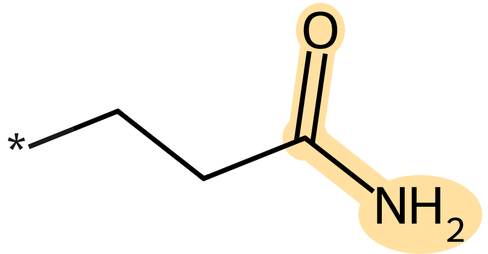
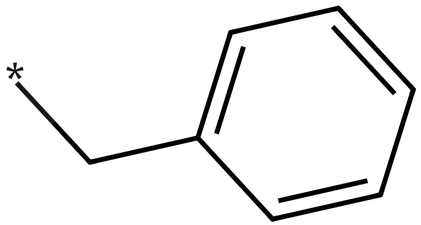
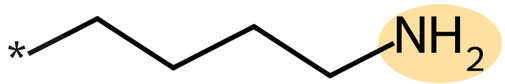
X A M A

Trace product

Which step creates the preference: activation, aminolysis, or both?

Literature product preference (regioselective oligoamide study, Fig. 7). Neutral identity drawings; molecular cause remains a hypothesis.

AS9: two mutations, two mechanistic questions

	WT SIDE CHAIN	MUTANT SIDE CHAIN	HYPOTHESIS TO TEST
PA66 diad **A215Q**			**Does the new amide reposition the substrate?**
PA55 diad **F249K**			**Does the new amine alter binding or packing?**

Next: matched structural comparisons + experimental controls

User-reported yield improvements; assay values pending. * = backbone attachment. Identity drawings do not specify assay protonation states.